

CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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MATHEMATICS

0580/42

Paper 4 (Extended)

May/June 2019

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials:

Electronic calculator

Geometrical instruments

Tracing paper (optional)

READ THESE INSTRUCTIONS FIRST

Write your centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer **all** questions.

If working is needed for any question it must be shown below that question.

Electronic calculators should be used.

If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

For π , use either your calculator value or 3.142.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

The total of the marks for this paper is 130.

This document consists of **19** printed pages and **1** blank page.

- 1 (a) The price of a newspaper increased from \$0.97 to \$1.13 .

Calculate the percentage increase.

..... % [3]

- (b) One day, the newspaper had 60 pages of news and advertisements.

The ratio number of pages of news : number of pages of advertisements = 5 : 7.

- (i) Calculate the number of pages of advertisements.

..... [2]

- (ii) Write the number of pages of advertisements as a percentage of the number of pages of news.

..... % [1]

- (c) On holiday Maria paid 2.25 euros for the newspaper when the exchange rate was \$1 = 0.9416 euros.
At home Maria paid \$1.13 for the newspaper.

Calculate the difference in price.

Give your answer in dollars, correct to the nearest cent.

\$ [3]

- (d) The number of newspapers sold decreases exponentially by $x\%$ each year.
Over a period of 21 years the number of newspapers sold decreases from 1 763 000 to 58 000.

Calculate the value of x .

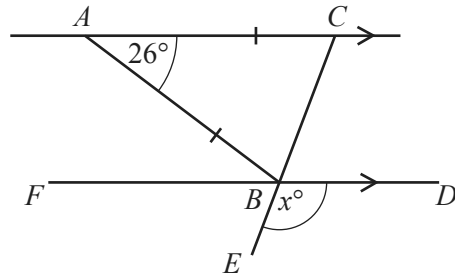
$x = \dots\dots\dots$ [3]

- (e) Every page of the newspaper is a rectangle measuring 43 cm by 28 cm, both correct to the nearest centimetre.

Calculate the upper bound of the area of a page.

$\dots\dots\dots \text{ cm}^2$ [2]

2 (a)

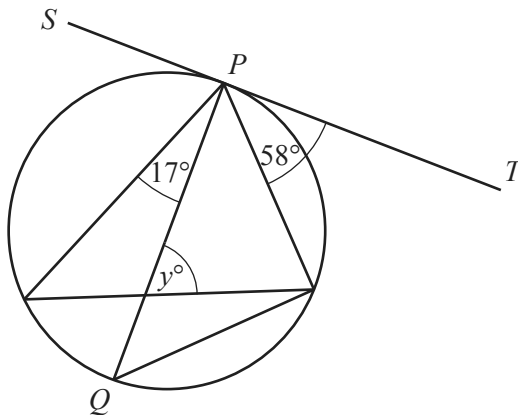
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AC is parallel to FBD , ABC is an isosceles triangle and CBE is a straight line.

Find the value of x .

$x = \dots\dots\dots$ [3]

(b)

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The diagram shows a circle with diameter PQ .
 SPT is a tangent to the circle at P .

Find the value of y .

$y = \dots\dots\dots$ [5]

3 The probability that Andrei cycles to school is r .

(a) Write down, in terms of r , the probability that Andrei **does not** cycle to school.

..... [1]

(b) The probability that Benoit **does not** cycle to school is $1.3 - r$.

The probability that both Andrei and Benoit **do not** cycle to school is 0.4 .

(i) Complete the equation in terms of r .

(.....) $\times$ (.....) = 0.4 [1]

(ii) Show that this equation simplifies to $10r^2 - 23r + 9 = 0$.

[3]

(iii) Solve by factorisation $10r^2 - 23r + 9 = 0$.

$r =$ or $r =$ [3]

(iv) Find the probability that Benoit **does not** cycle to school.

..... [1]

4 (a) The equation of a straight line is $2y = 3x + 4$.

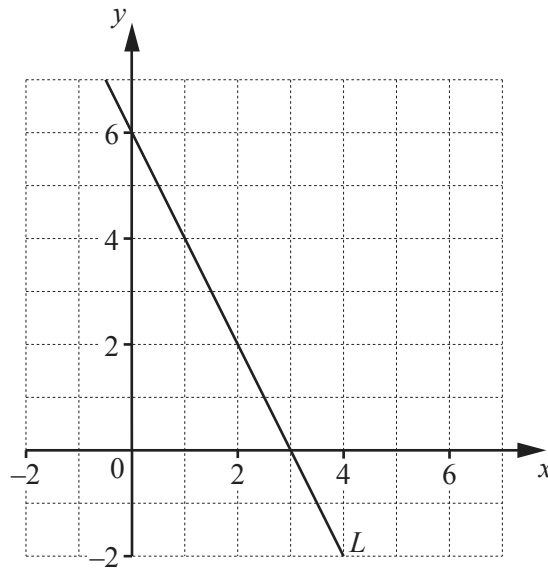
(i) Find the gradient of this line.

..... [1]

(ii) Find the co-ordinates of the point where the line crosses the y -axis.

(..... ,) [1]

(b) The diagram shows a straight line L .



(i) Find the equation of line L .

..... [3]

(ii) Find the equation of the line perpendicular to line L that passes through $(9, 3)$.

..... [3]

(c) A is the point $(8, 5)$ and B is the point $(-4, 1)$.

(i) Calculate the length of AB .

..... [3]

(ii) Find the co-ordinates of the midpoint of AB .

(..... ,) [2]

- 5 The table shows some values of $y = \frac{1}{2x} - \frac{x}{4}$ for $0.15 \leq x \leq 3.5$.

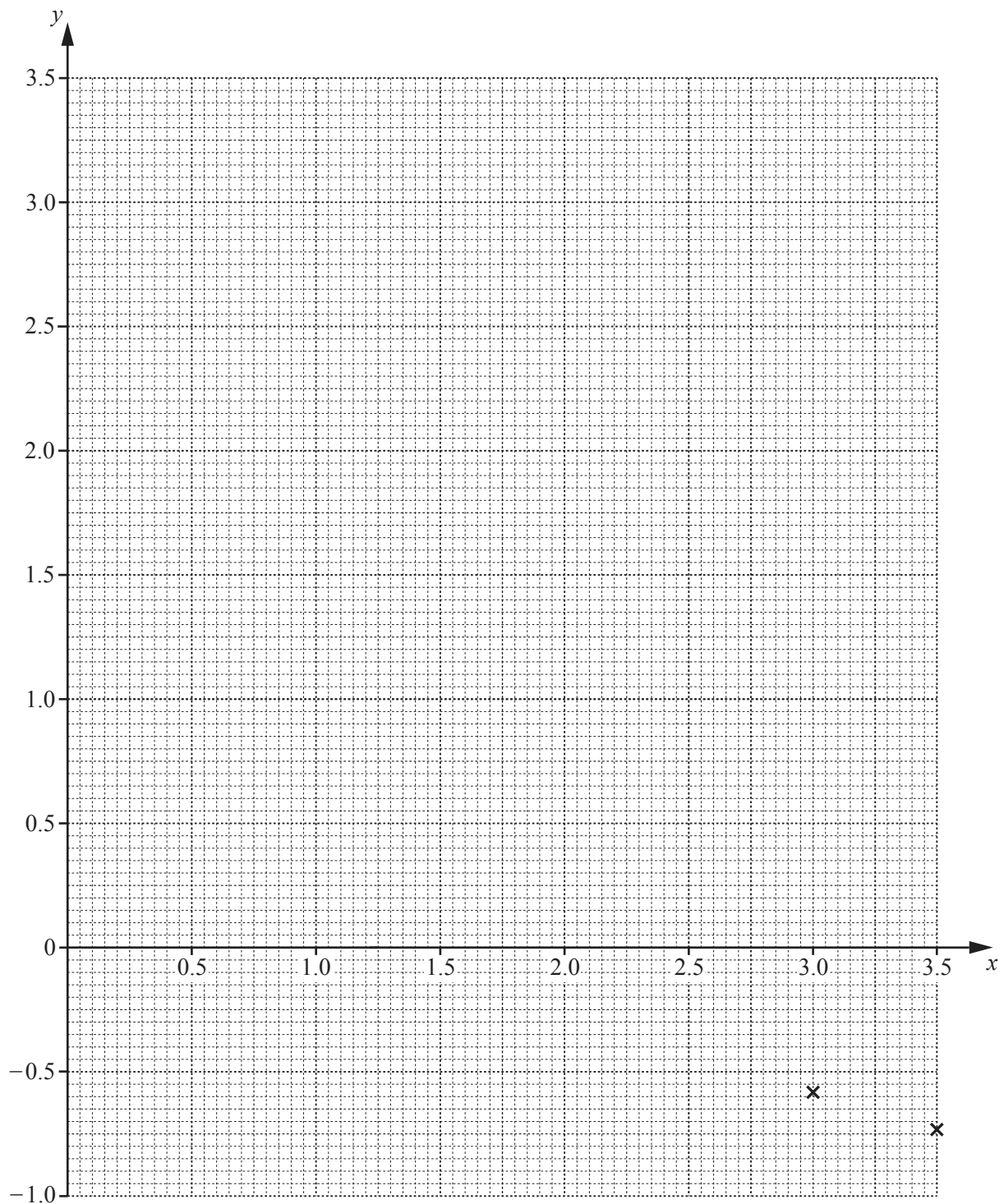
x	0.15	0.2	0.5	1	1.5	2	2.5	3	3.5
y	3.30		0.88		-0.04		-0.43	-0.58	-0.73

(a) Complete the table.

[3]

- (b) On the grid, draw the graph of $y = \frac{1}{2x} - \frac{x}{4}$ for $0.15 \leq x \leq 3.5$.

The last two points have been plotted for you.



[4]

- (c) Use your graph to solve the equation $\frac{1}{2x} - \frac{x}{4} = \frac{1}{2}$ for $0.15 \leq x \leq 3.5$.

$x = \dots\dots\dots$ [1]

- (d) (i) On the grid, draw the line $y = 2 - x$. [2]

- (ii) Write down the x co-ordinates of the points where the line $y = 2 - x$ crosses the graph of

$$y = \frac{1}{2x} - \frac{x}{4} \text{ for } 0.15 \leq x \leq 3.5.$$

$x = \dots\dots\dots$ and $x = \dots\dots\dots$ [2]

- (e) Show that the graph of $y = \frac{1}{2x} - \frac{x}{4}$ can be used to find the value of $\sqrt{2}$ for $0.15 \leq x \leq 3.5$.

[2]

- 6 (a) Expand and simplify.

$$(x+7)(x-3)$$

..... [2]

- (b) Factorise completely.

(i) $15p^2q^2 - 25q^3$

..... [2]

(ii) $4fg + 6gh + 10fk + 15hk$

..... [2]

(iii) $81k^2 - m^2$

..... [2]

- (c) Solve the equation.

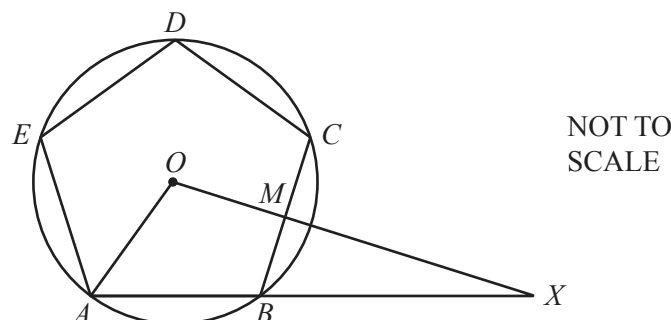
$$3(x-4) + \frac{x+2}{5} = 6$$

$x =$ [4]

- 7 (a) Show that each interior angle of a regular pentagon is 108° .

[2]

(b)



The diagram shows a regular pentagon $ABCDE$.
The vertices of the pentagon lie on a circle, centre O , radius 12 cm.
 M is the midpoint of BC .

- (i) Find BM .

$BM = \dots\dots\dots$ cm [3]

- (ii) OMX and ABX are straight lines.

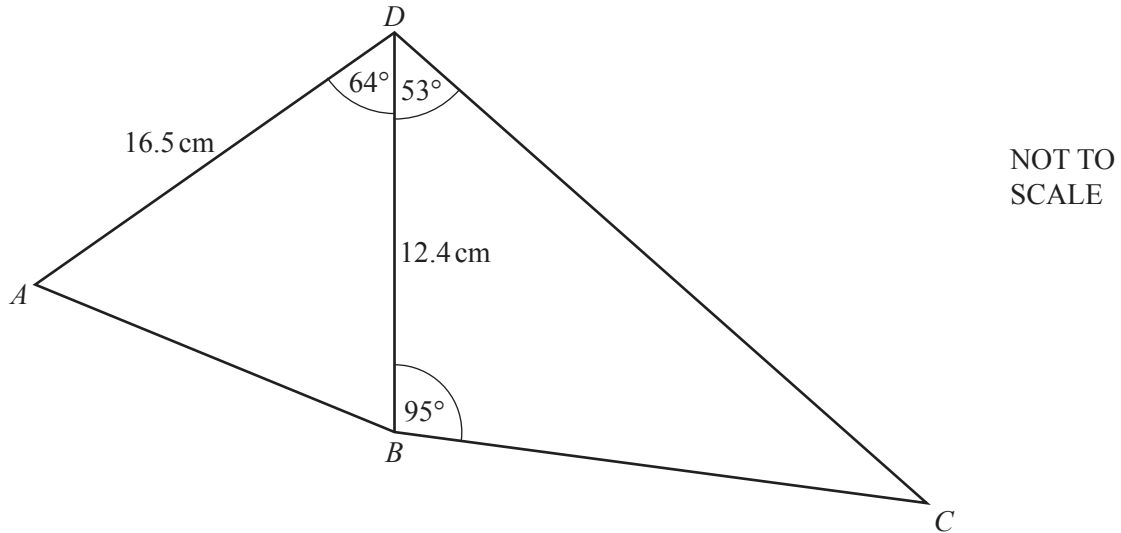
- (a) Find BX .

$BX = \dots\dots\dots$ cm [3]

- (b) Calculate the area of triangle AOX .

$\dots\dots\dots$ cm^2 [3]

8 (a)



The diagram shows two triangles ABD and BCD .
 $AD = 16.5$ cm and $BD = 12.4$ cm.
 Angle $ADB = 64^\circ$, angle $BDC = 53^\circ$ and angle $DBC = 95^\circ$.

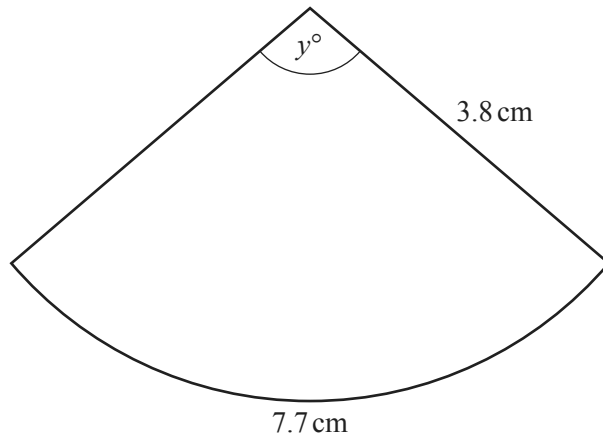
(i) Find AB .

$AB = \dots\dots\dots$ cm [4]

(ii) Find BC .

$BC = \dots\dots\dots$ cm [4]

(b)

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The diagram shows a sector of a circle of radius 3.8 cm.
The arc length is 7.7 cm.

(i) Calculate the value of y .

$y = \dots\dots\dots$ [2]

(ii) Calculate the area of the sector.

$\dots\dots\dots \text{ cm}^2$ [2]

- 9 100 students were each asked how much money, \$ m , they spent in one week. The frequency table shows the results.

Amount (\$ m)	$0 < m \leq 5$	$5 < m \leq 10$	$10 < m \leq 20$	$20 < m \leq 30$	$30 < m \leq 50$
Frequency	16	38	30	9	7

- (a) Calculate an estimate of the mean.

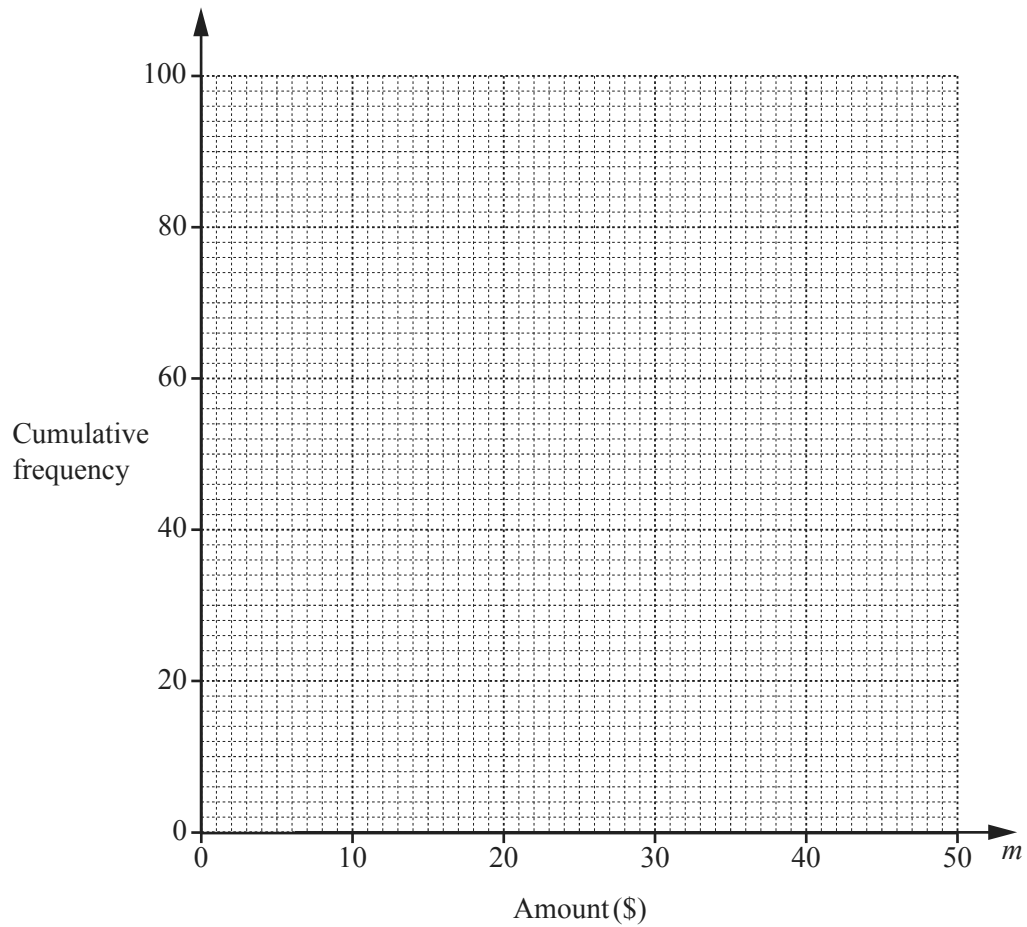
\$ [4]

- (b) Complete the cumulative frequency table below.

Amount (\$ m)	$m \leq 5$	$m \leq 10$	$m \leq 20$	$m \leq 30$	$m \leq 50$
Cumulative frequency	16				100

[2]

(c) On the grid, draw the cumulative frequency diagram.



[3]

(d) Use your cumulative frequency diagram to find an estimate for

(i) the median,

\$ [1]

(ii) the interquartile range,

\$ [2]

(iii) the number of students who spent more than \$25.

..... [2]

10 (a) The volume of a solid metal sphere is $24\,430\text{ cm}^3$.

(i) Calculate the radius of the sphere.

[The volume, V , of a sphere with radius r is $V = \frac{4}{3}\pi r^3$.]

..... cm [3]

(ii) The metal sphere is placed in an empty tank.

The tank is a cylinder with radius 50 cm, standing on its circular base.

Water is poured into the tank to a depth of 60 cm.

Calculate the number of litres of water needed.

..... litres [3]

(b) A different tank is a cuboid measuring 1.8 m by 1.5 m by 1.2 m.

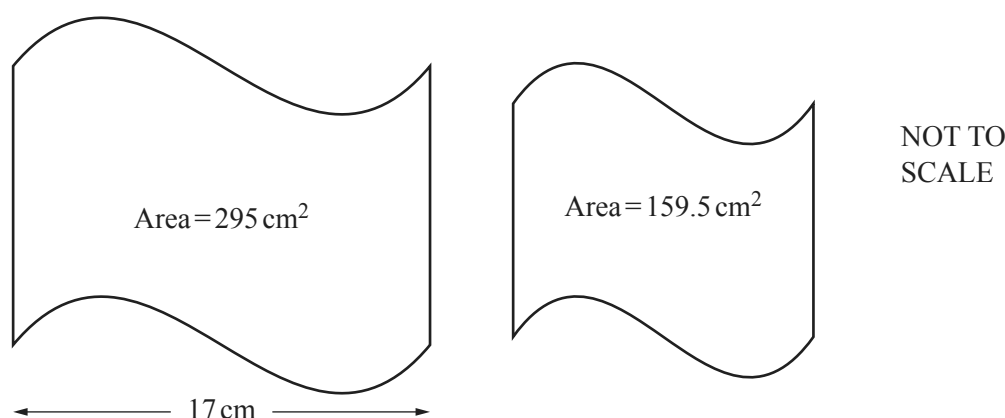
Water flows from a pipe into this empty tank at a rate of 200 cm^3 per second.

Find the time it takes to fill the tank.

Give your answer in hours and minutes.

..... hours minutes [4]

(c)

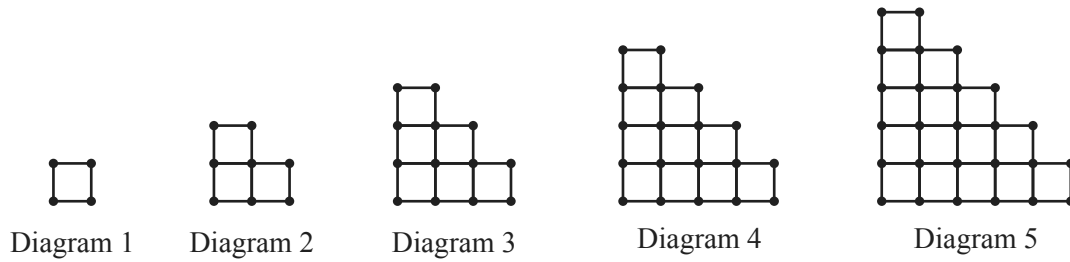


The diagram shows two mathematically similar shapes with areas 295 cm^2 and 159.5 cm^2 .
The width of the larger shape is 17 cm.

Calculate the width of the smaller shape.

..... cm [3]

11



The sequence of diagrams above is made up of small lines and dots.

(a) Complete the table.

	Diagram 1	Diagram 2	Diagram 3	Diagram 4	Diagram 5	Diagram 6
Number of small lines	4	10	18	28		
Number of dots	4	8	13	19		

[4]

(b) For Diagram n find an expression, in terms of n , for the number of small lines.

..... [2]

(c) Diagram r has 10 300 small lines.

Find the value of r .

$r =$ [2]

- (d) The number of dots in Diagram n is $an^2 + bn + 1$.

Find the value of a and the value of b .

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots [2]$$

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